

Tools/Technologies

Trello:

To keep the development process organised we are going to use Trello for our project management. This will allow us to collaborate, track tasks, set deadlines and monitor progress. Trello supports our agile methodology as it will easily display every sprint and tasks associated with them.

Plan is to use a kanban board for every sprint, this will keep things tidy as we will not have an overwhelming number of tasks (completed or not) on our screen at the same time. We will be migrating in progress or yet to begin functions / features to the next sprints board if we were unable to fit them in the sprint in appropriate time

Draw.io:

We will be using Draw.io for creating a visual representation of our database. This tool will allow us to map out tables, data types keys and relationships. It is also easy to collaborate and modify the layouts as it can be saved as a XML file which can be edited by anyone.

Visual Studio Code:

We all decided and agreed to use Visual Studio Code (VSC) as our IDE. It's super flexible while including helpful features such as debugging, git integration and tons of extension. VSC is both suitable for working on the frontend as well as the backend. Git Integration will let us collaborate and edit the code without an issue.

There are many extensions for us to use, for example PHP syntax for the backend & utility tools such as blade for the frontend. The ability to use the popular extensions will make our lives that bit easier as programmers on this project.

Another great use of VSC is the ability to open up terminal projects, this means that if we choose to store our project in terminal (or raptor) we can simply get the code into the IDE which is ten times better than using a light weight command line editor such as nano. Also VSC you can use terminal, for example to run our backend & frontend (npm run dev)

MySQL:

For our database management we've chosen MySQL as we will need to store and handle a fair amount of data. MySQL allows us to manage information such as reservations and menu items.

Our plan is to store the MySQL database on raptor, so we do not have to keep updating our locally hosted databases with the most up to date script as this can sometimes lead to complications in people forgetting to update their database via the latest script

One downside to using raptor is to remember to run the vpn before trying to run the project as without it we will not be able to access or database

- Apache, web server host (raptor)
- React.js, HTML, CSS, JavaScript, Bootstrap - Frontend
- Laravel & inertia / PHP framework – Backend

React:

We will be using React for the frontend, specifically for creating new pages as well as centralising specific reusable components. From what I can see React has some good libraries associated with it for creating easier frontends, making layouts for specific pages as well as making it easier to send & retrieve information from the backend. It is a javascript framework.

Laravel:

Laravel is a php frameworks which will give us a structure for the project, Laravel & php generally is used for backing tasks such as sending & retrieving information from the database. From my understanding Laravel will be used for page navigation as well as db calls, it should be pretty simple to establish a connection to the dragon db using this framework.

Inertia:

This is a middleware that is built into many of the pre configured Laravel project starter packs, this will make it easier to get the backend & frontend to communicate as they will be on separate servers, example dragon database on raptor & the project cloned from git onto our machines. This means we do not need to configure our own API as this achieves the same things while being built in.

Bootstrap:

This is a common CSS framework which most of use have used before, this will allow us to use the most up to date designs to ensure our pages look the part. This will make sure that our pages are all looking consistent no matter the device type the user is on.